Part V — Waves, Sound, and Light · Chapter 27

Mirrors and Lenses

How curved reflecting and refracting surfaces form images that magnify, invert, and focus the world

> *Astronomers using the Hubble Space Telescope look at galaxies billions of light-years away through a mirror 2.4 meters across. Surgeons guide instruments through incisions using fiber-optic endoscopes that relay images from inside the body. Smartphone cameras pack a multi-lens optical system thinner than a finger. A $2 magnifying glass and a $10 000 camera lens obey exactly the same equations. The physics that makes all of this possible — the geometry of how curved mirrors and lenses redirect light to form images — was fully worked out by Newton, Huygens, and Gauss in the 17th and 18th centuries. This chapter develops those tools completely, from the geometry of ray tracing through the mirror and lens equations that make optical design systematic.*

27.1 Why Optical Systems Form Images

An image is formed when rays of light from a single point on an object are redirected — by reflection or refraction — so that they converge at, or appear to diverge from, a single point in space that is different from the object location. That new convergence (or apparent divergence) point is the image of the object point.

A flat mirror does this: rays from a point on your face reflect off the mirror and diverge on the observer's side as if they came from a point behind the mirror. That point is the image. A camera lens does this too: rays from a point on the subject converge to a point on the film or sensor, forming a real image.

Whether the image is real or virtual, upright or inverted, magnified or reduced depends on the geometry of the optical element and the position of the object. Three-dimensional images of extended objects are formed point-by-point: every point on the object has a corresponding image point, and the collection of all image points constitutes the image of the full object.

The tools we use to find image properties are ray tracing (drawing specific predictable rays to locate images geometrically) and the mirror/lens equations (algebraic relationships that give image distance and magnification from object distance and focal length). Both methods are essential, and they always agree when applied correctly.

27.2 Review of Reflection and Refraction Principles

Chapter 26 established the two laws governing how light behaves at boundaries:

Law of Reflection: $\theta_r = \theta_i$, both measured from the normal. For a curved mirror, the normal at any point is the line connecting that point to the center of curvature of the mirror surface.

Snell's Law of Refraction: $n_1 \sin\theta_1 = n_2 \sin\theta_2$. For a lens, light refracts at two surfaces (the entry and exit surfaces), and the combined effect is to converge or diverge the beam.

In this chapter we apply these laws to curved surfaces and build up the complete theory of image formation for mirrors and lenses. The key simplification we use throughout is the paraxial approximation: we restrict our analysis to rays that travel close to and nearly parallel to the optical axis (the line through the centers of curvature of the mirror or lens). In this regime, the trigonometric functions simplify and the equations become linear and manageable.

> **Key Point:** All mirror and lens formulas in this chapter assume the paraxial approximation. For wide beams or large-aperture optics, higher-order effects (aberrations) cause the formulas to break down. Aberration correction is the central problem of advanced optical design.

27.3 Plane Mirror Images Revisited

Before addressing curved mirrors, it is worth revisiting the plane mirror to establish notation and sign conventions that will extend to curved surfaces. For a plane mirror:

- Object distance d_o: positive, measured from mirror to object in front of the mirror.
- Image distance d_i: negative for a plane mirror (image is behind the mirror, on the virtual side).
- Focal length f: undefined for a plane mirror (or treated as $f \to \infty$).
- Magnification $m = -d_i/d_o = -(-d_o)/d_o = +1$ (image same size, upright).

The sign convention used throughout this chapter is the Cartesian sign convention: distances measured in the direction of the incoming light are positive; distances measured opposite to the incoming light are negative. For mirrors, the incoming and outgoing light are on the same side, so virtual images (behind the mirror, opposite to the incoming light) have negative d_i.

27.4 Concave and Convex Mirrors

A curved mirror is a section of a sphere. The center of curvature C is the center of the sphere of which the mirror is a part. The radius of curvature R is the radius of that sphere. The focal point F is the point at which parallel rays converge (for a concave mirror) or appear to diverge from (for a convex mirror) after reflection. The focal length f = R/2.

A concave mirror (converging mirror) curves inward toward the incoming light, like the inside of a bowl. Parallel rays incident on a concave mirror converge to the focal point F in front of the mirror after reflection. Concave mirrors can form both real and virtual images depending on object position.

A convex mirror (diverging mirror) curves outward away from the incoming light, like the outside of a sphere. Parallel rays incident on a convex mirror diverge after reflection, appearing to come from the focal point F behind the mirror. Convex mirrors always form virtual, upright, reduced images. They produce a wide field of view, making them ideal for security mirrors, car side mirrors, and rear-view mirrors in stores.

Mirror Type	Curvature	Sign of f	Image Type (typical)	Common Use
Concave (converging)	Curves inward	$f > 0$ (positive)	Real or virtual	Telescopes, makeup mirrors, satellite dishes
Convex (diverging)	Curves outward	$f < 0$ (negative)	Always virtual, reduced	Car side mirrors, security mirrors
Plane	Flat	$f = \infty$	Always virtual, same-size	Bathroom mirrors, periscopes

Table 27.1 *Summary of mirror types and their basic properties.*

27.5 Principal Rays for Mirror Ray Tracing

Ray tracing uses three specific predictable rays to locate images geometrically. You need only two of the three to find the image; the third serves as a check.

Ray 1 — Parallel Ray

A ray parallel to the optical axis strikes the mirror and reflects through the focal point F (for a concave mirror) or reflects as if it came from F (for a convex mirror, where F is behind the mirror).

Ray 2 — Focal Ray

A ray passing through (or directed toward) the focal point F strikes the mirror and reflects parallel to the optical axis. This is the time-reversal of Ray 1.

Ray 3 — Center Ray

A ray directed toward (or through) the center of curvature C strikes the mirror and reflects straight back along the same path, because it hits the surface perpendicularly (normal passes through C). The center of curvature is at distance R = 2f from the mirror.

To trace an image: draw Rays 1 and 2 (or any two) from the top of the object. For a concave mirror, the point where the reflected rays cross (in front of the mirror) is the real image location. If the reflected rays diverge, extend them backward behind the mirror; where they appear to cross is the virtual image. For a convex mirror, reflected rays always diverge; extend them backward to find the virtual image.

> **The Rule:** Real images form in front of a mirror (where reflected rays actually cross). Virtual images form behind a mirror (where reflected rays appear to come from when extended backward). Real images can be projected on a screen; virtual images cannot.

27.6 Mirror Image Formation

The nature of the image formed by a concave mirror depends on where the object is placed relative to the focal point F and the center of curvature C (at distance 2f).

Object Location	Image Location	Image Type	Orientation	Size
Beyond C (d_o > 2f)	Between F and C	Real	Inverted	Reduced
At C (d_o = 2f)	At C	Real	Inverted	Same size
Between F and C	Beyond C (d_o > 2f)	Real	Inverted	Enlarged

At F ($d_o = f$)	At infinity	—	—	— (parallel reflected rays)
Inside F ($d_o < f$)	Behind mirror	Virtual	Upright	Enlarged

Table 27.2 *Image properties for a concave mirror as a function of object position.*

For a convex mirror, the image is always virtual, upright, and reduced, regardless of object position. The image forms between the mirror surface and the focal point behind the mirror. As the object moves closer to the mirror, the image grows larger but remains behind the mirror.

The make-up or shaving mirror is a concave mirror: the face is placed inside the focal length, producing a virtual, upright, enlarged image. A headlight reflector is also concave: the light source (bulb) is placed at the focal point, producing a parallel reflected beam (the headlight).

27.7 The Mirror Equation

The mirror equation relates object distance d_o, image distance d_i, and focal length f algebraically:

$$1/f = 1/d_o + 1/d_i$$

Mirror equation. f = focal length; d_o = object distance; d_i = image distance. All measured from the mirror vertex.

Sign conventions (Cartesian, for mirrors): distances measured in front of the mirror (where light reflects) are positive. Distances behind the mirror are negative. Focal length $f > 0$ for concave mirrors; $f < 0$ for convex mirrors.

Step 1: Identify: f (from mirror type and radius), d_o (object distance, always positive for real objects).

Step 2: Solve: $1/d_i = 1/f - 1/d_o$, then $d_i = 1/(1/f - 1/d_o)$.

Step 3: Interpret: $d_i > 0$ → real image in front of mirror. $d_i < 0$ → virtual image behind mirror.

Example — Concave mirror, object beyond F $f = +20$ cm, $d_o = 60$ cm. $1/d_i = 1/20 - 1/60 = 3/60 - 1/60 = 2/60$. $d_i = 30$ cm (positive → real, in front of mirror).

Example — Convex mirror $f = -30$ cm, $d_o = 20$ cm. $1/d_i = 1/(-30) - 1/20 = -1/30 - 1/20 = -2/60 - 3/60 = -5/60$. $d_i = -12$ cm (negative → virtual, behind mirror).

27.8 Magnification in Mirrors

The lateral magnification m is the ratio of the image height h_i to the object height h_o. It is also given by the ratio of the image distance to the object distance with a sign:

$$m = h_i / h_o = - d_i / d_o$$

Lateral magnification for mirrors. $m > 0 \rightarrow$ upright image. $m < 0 \rightarrow$ inverted image. $|m| > 1 \rightarrow$ enlarged. $|m| < 1 \rightarrow$ reduced.

The sign of m tells you the orientation: positive magnification means the image is upright (same orientation as the object); negative magnification means inverted. The magnitude $|m|$ tells you the size ratio.

Example — Magnification from previous example Concave mirror, $d_o = 60$ cm, $d_i = 30$ cm. $m = -d_i/d_o = -30/60 = -0.5$. Image is inverted (negative) and half the size of the object ($|m| = 0.5 < 1 \rightarrow$ reduced).

Example — Convex mirror magnification $d_o = 20$ cm, $d_i = -12$ cm. $m = -(-12)/20 = +0.60$. Image is upright (positive) and 60% of object size (reduced).

A concave mirror used as a makeup mirror has $d_o < f$, giving d_i negative (virtual image behind mirror) and $m > +1$ (upright, enlarged). The magnification of a concave mirror with $f = 15$ cm and $d_o = 10$ cm: $1/d_i = 1/15 - 1/10 = 2/30 - 3/30 = -1/30$, so $d_i = -30$ cm. $m = -(-30)/10 = +3$. The image is upright and 3× the object size.

27.9 Converging and Diverging Lenses

A lens is a transparent optical element with at least one curved surface that refracts light to form images. Unlike mirrors (which work by reflection), lenses work by refraction at two surfaces. The net effect of the two refractions determines whether the lens converges or diverges light.

A converging (convex) lens is thicker at the center than at the edges. It refracts parallel rays so they converge to a real focal point F on the far side of the lens. A simple converging lens has a positive focal length ($f > 0$). Examples: magnifying glasses, camera lenses, the human eye's crystalline lens, telescope objectives.

A diverging (concave) lens is thinner at the center than at the edges. It refracts parallel rays so they diverge, appearing to come from a virtual focal point on the same side as the incoming light. A diverging lens has a negative focal length ($f < 0$). Examples: corrective lenses for nearsightedness, the eyepiece element in some telescopes, wide-angle camera elements.

Lens Type	Shape	Sign of f	Image (typical)	Common Use
Converging (convex)	Thicker center	$f > 0$	Real or virtual	Camera, magnifier, eye
Diverging (concave)	Thinner center	$f < 0$	Always virtual	Glasses for myopia, telescope eyepiece

Table 27.3 *Converging and diverging lens properties.*

The focal length of a lens depends on the index of refraction of the glass and the radii of curvature of its two surfaces (the Lensmaker's Equation, not derived here). For a given glass, stronger curvatures give shorter focal lengths (more powerful lenses). Lens power, measured in diopters (D), is defined as $P = 1/f$ where f is in meters. A lens with $f = 0.25$ m has power 4 D.

27.10 Principal Rays for Lens Ray Tracing

The same three-ray tracing technique used for mirrors applies to lenses, with modifications for the two-sided nature of lenses. Two of the three rays uniquely locate the image; the third checks consistency.

Ray 1 — Parallel Ray

A ray parallel to the optical axis refracts through the lens and passes through the far focal point F_2 (for a converging lens) or appears to come from the near focal point F_1 (for a diverging lens).

Ray 2 — Focal Ray

A ray passing through the near focal point F_1 (converging) or directed toward the far focal point F_2 (diverging) emerges from the lens parallel to the optical axis. Time-reversal of Ray 1.

Ray 3 — Center Ray

A ray passing through the center of the lens continues straight through without bending (the two surfaces at the center are approximately parallel, and any deflection cancels out for a thin lens).

For a thin converging lens: draw Rays 1 and 2 from the top of the object. Where they cross on the far side of the lens is the real image. If the object is inside the focal length, the refracted rays diverge; extend them backward to find the virtual image on the same side as the object.

For a thin diverging lens: rays always diverge after the lens. Extend them backward to the same side as the object to find the virtual image, which is always between the lens and the object-side focal point.

27.11 Lens Image Formation

The image properties for a converging lens depend on object position relative to the focal length, in exact analogy with the concave mirror.

Object Location	Image Location	Type	Orientation	Size
Beyond 2f	Between f and 2f (far side)	Real	Inverted	Reduced
At 2f	At 2f (far side)	Real	Inverted	Same size
Between f and 2f	Beyond 2f (far side)	Real	Inverted	Enlarged
At f	At infinity	—	—	Parallel beam
Inside f (d_o < f)	Same side as object	Virtual	Upright	Enlarged

Table 27.4 *Image properties for a converging lens as a function of object distance.*

For a diverging lens, the image is always virtual, upright, and reduced, located between the lens and the object-side focal point — analogous to the convex mirror. A diverging lens cannot form a real image of a real object.

27.12 The Lens Equation

The thin lens equation is identical in form to the mirror equation:

$$1/f = 1/d_o + 1/d_i$$

Thin lens equation. Same form as the mirror equation. Sign conventions differ slightly (see below).

Sign conventions for lenses (Cartesian): d_o is positive for real objects (light comes in from the left, object is on the left). d_i is positive for real images (formed on the far side of the lens, to the right) and negative for virtual images (formed on the same side as the object, to the left). f is positive for converging lenses, negative for diverging lenses.

> **Example — Converging lens, real image** $f = +15$ cm, $d_o = 45$ cm. $1/d_i = 1/15 - 1/45 = 3/45 - 1/45 = 2/45$. $d_i = 22.5$ cm (positive → real image on far side).
>
> **Example — Converging lens, virtual image (magnifier)** $f = +10$ cm, $d_o = 6$ cm. $1/d_i = 1/10 - 1/6 = 3/30 - 5/30 = -2/30$. $d_i = -15$ cm (negative → virtual, on same side as object).
>
> **Example — Diverging lens** $f = -20$ cm, $d_o = 30$ cm. $1/d_i = 1/(-20) - 1/30 = -3/60 - 2/60 = -5/60$. $d_i = -12$ cm (negative → virtual, on same side as object).

27.13 Magnification in Lenses

The lateral magnification formula for lenses is identical to that for mirrors:

$$m = h_i / h_o = -d_i / d_o$$

Lateral magnification for lenses. $m > 0$ → upright. $m < 0$ → inverted. $|m| > 1$ → enlarged. $|m| < 1$ → reduced.

> **Example — Magnification of the magnifier example** $d_o = 6$ cm, $d_i = -15$ cm. $m = -(-15)/6 = +2.5$. Upright (positive) and 2.5× enlarged.

A camera images a person 3 m tall standing 5 m from a lens with $f = 50$ mm $= 0.050$ m. $1/d_i = 1/0.050 - 1/5 = 20 - 0.2 = 19.8$. $d_i = 1/19.8 = 0.0505$ m $= 50.5$ mm. $m = -0.0505/5 = -0.0101$. The image is inverted (negative m) and about 1% of the object's height: 3 m × 0.0101 = 30.3 mm on the sensor. This is a realistic camera image size.

27.14 Compound Optical Instruments (Introductory)

Most practical optical instruments use more than one lens or mirror. In a compound system, the image formed by the first element becomes the object for the second element. This is the key principle: the image of one element is the object of the next.

The simple microscope (magnifying glass) uses a single converging lens. A compound microscope uses two converging lenses: the objective (short focal length, close to the specimen) forms a real, enlarged, inverted image inside the instrument, and the eyepiece (ocular) acts as a magnifying glass to further enlarge that intermediate image for the eye. The total magnification is the product of the individual magnifications of the two lenses.

The refracting telescope uses a large-focal-length objective to collect and focus light from distant objects (forming a small, real, inverted image at its focal plane) and an eyepiece of short focal length to magnify that image. Angular magnification = f_objective / f_eyepiece. Increasing f_objective or decreasing f_eyepiece both increase magnification.

The reflecting telescope replaces the objective lens with a large concave mirror, avoiding chromatic aberration (which plagues refracting objectives) and making very large apertures practical. The Hubble Space Telescope uses a 2.4 m concave mirror as its primary; the James Webb Space Telescope uses a 6.5 m segmented mirror array.

The human eye is itself a converging optical system. The cornea provides most of the refractive power (about +43 D); the crystalline lens adds adjustable power (about +19–26 D, controlled by ciliary muscles). Total eye power is about 60–70 D, focusing images on the retina ($d_i \approx 17$ mm). Eyeglasses and contact lenses correct the eye's focal length when the cornea or lens does not have the right curvature or power for the eye's dimensions.

> **Historical Note:** Galileo Galilei constructed his first telescope in 1609 after hearing of Hans Lippershey's Dutch design. Within a year, using a telescope magnifying about 30×, he had observed the moons of Jupiter, the phases of Venus, the mountains of the Moon, and sunspots — all of which contradicted the Ptolemaic geocentric model and supported Copernican heliocentrism.

27.15 Common Mistakes in Mirror and Lens Analysis

- Using the wrong sign convention. Choose one convention (Cartesian is standard in most textbooks) and apply it consistently. The most common error is mixing positive and negative d_i interpretations from different conventions.
- Forgetting that f is negative for convex mirrors and diverging lenses. If you solve the equation and get a physically unreasonable result (real image on the wrong side of a diverging element), check the sign of f.
- Misidentifying which side of the lens/mirror is positive. For mirrors: positive is in front (where light goes after reflection). For lenses: positive is on the far side from the object (where transmitted light goes). Virtual images are on the negative side in both cases.
- Confusing object position and image position in compound systems. In a two-lens system, you must find the image from the first lens, use that as the object

for the second lens (with appropriate sign for distance), and then apply the lens equation again.

- Using incorrect principal ray directions for convex mirrors and diverging lenses. Ray 1 (parallel) reflects or refracts as if coming from (not going to) the focal point. Ray 2 (directed toward F) emerges parallel. These are reversed compared to the concave/converging case.
- Computing magnification without the negative sign in $m = -d_i/d_o$. The negative sign is not optional; it is what gives the image orientation information. Forgetting it gives a magnification with the wrong sign, predicting an upright image when the image is actually inverted, or vice versa.

Common Mistake: For the mirror and lens equations to work, use consistent units throughout (all in cm, or all in m). Never mix cm and m in the same equation.

Chapter Summary

Optical images are formed when reflected or refracted rays converge at (real image) or appear to diverge from (virtual image) a point in space. Ray tracing and the mirror/lens equations are equivalent, complementary tools for finding image properties.

- Concave mirrors ($f > 0$) can form real or virtual images; convex mirrors ($f < 0$) always form virtual, upright, reduced images.
- Three principal rays: (1) parallel → through/from F; (2) through/from F → parallel; (3) through center → straight back (mirrors) or straight through (lens center).
- Mirror/lens equation: $1/f = 1/d_o + 1/d_i$. Lateral magnification: $m = -d_i/d_o$. Same equations for both mirrors and thin lenses.
- Converging lenses ($f > 0$) can form real or virtual images; diverging lenses ($f < 0$) always form virtual, upright, reduced images.
- Sign convention: real images have positive d_i; virtual images have negative d_i. $f > 0$ for concave mirrors and converging lenses; $f < 0$ for convex mirrors and diverging lenses.
- In compound instruments, the image of one element is the object for the next. Total magnification is the product of individual magnifications.
- The human eye has total power ~60–70 D; glasses correct its focal length. Microscopes amplify by objective then eyepiece. Telescopes amplify by f_{obj}/f_{eye}.

Key Terms

image: The point (or collection of points) to which reflected or refracted rays converge (real image) or from which they appear to diverge (virtual image), forming a representation of the object.

real image: An image formed where reflected or refracted rays actually converge; can be projected onto a screen; $d_i > 0$.

virtual image: An image formed where rays appear to diverge when extended backward; cannot be projected; $d_i < 0$.

optical axis: The line of symmetry of a mirror or lens system, passing through the center of curvature and focal points.

center of curvature (C): The center of the sphere of which a curved mirror surface is a part; located at distance R from the mirror vertex.

radius of curvature (R): The radius of the sphere of which a curved mirror is a part; $R = 2f$.

focal point (F): The point at which parallel rays converge (converging mirror/lens) or appear to diverge from (diverging mirror/lens) after reflection or refraction.

focal length (f): The distance from the mirror or lens to the focal point; $f = R/2$ for mirrors; $f > 0$ for converging, $f < 0$ for diverging.

concave mirror: A mirror that curves inward; has positive focal length; can form real and virtual images.

convex mirror: A mirror that curves outward; has negative focal length; always forms virtual, upright, reduced images.

converging lens: A lens thicker at the center; $f > 0$; converges parallel rays to a real focal point; can form real or virtual images.

diverging lens: A lens thinner at the center; $f < 0$; diverges parallel rays; always forms virtual, upright, reduced images.

mirror/lens equation: $1/f = 1/d_o + 1/d_i$; relates object distance, image distance, and focal length.

lateral magnification (m): $m = h_i/h_o = -d_i/d_o$; describes image size and orientation relative to the object.

paraxial approximation: The restriction to rays traveling near and nearly parallel to the optical axis, which linearizes the mirror/lens equations.

diopter (D): Unit of lens power; $P = 1/f$ (f in meters). A lens with $f = 0.5$ m has $P = 2$ D.

compound microscope: A two-lens instrument using an objective of short focal length and an eyepiece; total magnification = m_obj × m_eye.

reflecting telescope: A telescope using a large concave mirror as the objective; avoids chromatic aberration; practical for very large apertures.

Concept Review Questions

Answer in complete sentences. No calculations required unless specified.

1. Explain the difference between a real image and a virtual image. Why can a real image be projected on a screen while a virtual image cannot?
2. A convex mirror always produces a virtual, upright, reduced image regardless of object position. Explain why, using the behavior of the three principal rays.
3. A concave mirror produces an enlarged image when the object is inside the focal length, and a reduced image when the object is far beyond the focal length. Explain this qualitatively using the mirror equation.
4. Why does a diverging lens always produce a virtual image of a real object? Trace the three principal rays for a diverging lens to support your answer.
5. A camera takes a sharp photo of a person 5 m away. The photographer then walks toward a flower 0.2 m away. What must the photographer adjust to take a sharp photo of the flower? Explain using the lens equation.
6. Why does a reflecting telescope avoid chromatic aberration while a refracting telescope does not? What type of aberration do both share?
7. Explain why the magnification formula has a negative sign: m = −d_i/d_o. What does the sign of m tell you, and what does its magnitude tell you?
8. A person is nearsighted (myopic): their eye focuses parallel rays in front of the retina. What type of corrective lens (converging or diverging) is needed? Explain why.

Ray Tracing Exercises

For each case, describe (do not draw) where the image forms, whether it is real or virtual, upright or inverted, and enlarged or reduced. Confirm with the mirror/lens equation.

9. Concave mirror, $f = 20$ cm, object at $d_o = 60$ cm.
10. Concave mirror, $f = 20$ cm, object at $d_o = 10$ cm.
11. Convex mirror, $f = -25$ cm, object at $d_o = 40$ cm.
12. Converging lens, $f = 15$ cm, object at $d_o = 40$ cm.
13. Converging lens, $f = 15$ cm, object at $d_o = 8$ cm (inside f).
14. Diverging lens, $f = -18$ cm, object at $d_o = 30$ cm.

Mirror Equation Problems

15. A concave mirror has a radius of curvature of 40 cm. An object is placed 60 cm from the mirror. Find (a) the focal length, (b) the image distance, (c) the magnification, and (d) describe the image.
16. A convex mirror in a store has focal length −50 cm. A shopper stands 2.0 m from the mirror. Find the image distance and magnification.
17. A concave mirror forms a real image twice the size of the object. The image is 30 cm from the mirror. Find the object distance and focal length.
18. A dentist's mirror must form an upright image 4.0 times larger than the object when held 1.5 cm from a tooth. Find the focal length and radius of curvature needed.
19. A concave mirror ($f = 25$ cm) is used as a solar collector. At what distance from the mirror will the Sun's image form? (Treat the Sun as an object at $d_o = \infty$.)

Lens Equation Problems

20. A converging lens has focal length 10 cm. An object is placed at (a) 30 cm, (b) 10 cm, (c) 5 cm. Find image distance and magnification in each case.
21. A diverging lens has focal length −15 cm. An object is 25 cm from the lens. Find image distance, magnification, and describe the image.
22. A camera lens has $f = 55$ mm. It must focus a subject at 2.0 m onto the sensor. How far from the lens must the sensor be placed?

23. A 2× magnifying glass ($m = +2$ when used normally) has the object inside the focal length. If $d_o = 6$ cm, find the focal length.
24. A nearsighted person needs glasses with power −2.5 D. Find the focal length. What does the negative power tell you about the lens type?

Optical Instrument Applications

25. A compound microscope has an objective of focal length 4.0 mm and an eyepiece of focal length 25.0 mm. The objective forms an image 160 mm beyond the objective. Find (a) the image distance from the objective, (b) the magnification of the objective, and (c) the angular magnification of the eyepiece ($m_{eye} = 25$ cm / f_{eye} for a standard near point of 25 cm). What is the total magnification?
26. A refracting telescope has an objective with $f_{obj} = 900$ mm and an eyepiece with $f_{eye} = 25$ mm. Find the angular magnification. If the Moon subtends 0.5° to the naked eye, what angle does the Moon subtend through this telescope?
27. A person with a near point of 30 cm (requires objects at 30 cm for comfortable close vision) uses a magnifying glass of $f = 6.0$ cm. The image forms at the near point ($d_i = -30$ cm). Find d_o and the magnification.
28. A patient has a far point (maximum distance of clear vision) of 1.5 m. Find the power (in diopters) of the corrective contact lens needed to allow the patient to see clearly at infinity.

Chapter 27 Checkpoint

Before moving on to Chapter 28, confirm you can do each of the following.

I can...	See Section
Explain what an image is and distinguish real from virtual images	27.1
Apply the sign convention for d_o, d_i, and f for both mirrors and lenses	27.2–27.3

Distinguish concave from convex mirrors and state the sign of f for each	27.4
Draw the three principal rays for a mirror and locate the image	27.5
Predict image properties (real/virtual, orientation, size) from object position	27.6
Apply the mirror equation 1/f = 1/d_o + 1/d_i	27.7
Calculate lateral magnification and interpret its sign and magnitude	27.8
Distinguish converging and diverging lenses and state the sign of f for each	27.9
Draw the three principal rays for a lens and locate the image	27.10
Predict lens image properties from object position relative to f	27.11
Apply the thin lens equation and find d_i and m	27.12–27.13
Explain how compound microscopes and telescopes produce magnification	27.14
Identify and avoid common errors in mirror and lens equation problems	27.15

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